Observer-Patch Holography Cosmology Data and Likelihood Contracts

Provenance, Comparability, and Claim Boundaries

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Abstract

We specify data and likelihood contracts for conditional Observer-Patch Holography (OPH) cosmology. Each comparison binds a finite source, compatible transfer equations, declared datasets and covariance, nuisance parameters, and a common observable projection. The contracts distinguish source-derived quantities, imported inputs and fitted parameters, and account for sample overlap and shared calibration. They give a reproducible interface from screen and dark-sector models to cosmological data. A retrospective comparison ledger with an independent verifier and machine-checked brackets records where the receipt-conditional values land on Planck, ACT, SPT, BICEP/Keck, DESI and SPARC data: the tilt, running, tensor, isocurvature, non-Gaussianity and curvature rows are consistent, the fixed-capacity equation of state is in tension with DESI DR2, and every row is on seen data.

1 Comparison Classes

Every reported number belongs to one of four classes:

1. **source-derived**: all physical inputs follow from the declared OPH construction without using the comparison data;
2. **imported**: established background or microphysical inputs are declared explicitly;
3. **fit**: one or more parameters are inferred from the same data;
4. **diagnostic**: the calculation tests shape, scale, code behavior, or an interface without supporting a physical OPH claim.

The cosmology comparisons in scope are diagnostic or imported. No source-derived joint cosmology likelihood exists. The retrospective comparison ledger of Section 6 classifies every recorded comparison by these classes and by whether the compared value is shared with the standard baseline.

The spacetime construction provides a flat causal-order and normalized count-volume limit for conservative record populations with a specified neighbour-read law and layer duration. With complete authenticated ancestry and anchors, the interval-count ratio $(|I|/|J|)^{1/4}$ approaches the proper-time ratio of interior timelike diamonds, relative to a nondegenerate reference interval in the same history. The readout needs no coordinates, timestamps or density; its convergence uses the supplied population and law.

A separate cone-selection criterion requires a nonzero closed convex pointed cone invariant under the proper icosahedral rotations and all rapidities of one boost axis. Its operational action on histories, interventions and clock readouts is additional to the algebraic source frame. In a likelihood comparison, that hypothesis, the event law and its physical calibration count as model information, alongside the background, conserved stress, initial conditions and matter assumptions. A geometric count clock by itself does not define the cosmological transfer or observable likelihood.

2 Required Model Information

A physical comparison must state:

1. the covariant action and source map;
2. the background solution and species content;
3. the gauge-invariant initial conditions;
4. the perturbation, collision, and exchange equations;
5. the observable projection and numerical tolerances;
6. every external input and fitted parameter.

For the modular-charge dark-sector proposal, conserved charge in a supplied comoving volume gives the homogeneous dilution law $\rho_A \propto a^{-3}$. With the separate no-exchange continuity equation it gives a pressureless background. The abundance, relativistic stress, initial perturbations, lensing map, clusters, and nonlinear structure equations are not supplied here.

3 Required Data Information

A comparison must identify the data release, selection, masks, covariance, calibration model, nuisance parameters, priors, and combination rule. Dataset overlap and shared calibration must be included. A diagonal sum is not a joint likelihood when the covariance is non-diagonal.

The model and data layers are separate. Measured values that select a source formula, normalization, branch, or applicability domain are inputs to that comparison rather than OPH outputs.

4 Observable Contracts

Observable family	Required theory output	Required comparison object
Cosmic microwave background (CMB)	C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE}, lensing, foreground model	map or band-power likelihood with covariance and nuisance model
Baryon acoustic oscillations (BAO) and supernovae	$H(z)$, $D_A(z)$, luminosity distance, sound horizon	survey likelihood with calibration covariance
Weak lensing	metric potentials, nonlinear matter power, intrinsic alignments	tomographic likelihood with masks and covariance
Growth and redshift-space distortions (RSD)	$P_m(k, z)$ and $f\sigma_8(z)$	windowed survey likelihood with cross-bin covariance
Clusters	mass function, selection, observable–mass map, lensing calibration	count and calibration likelihood
Galaxies	disk dynamics, gas and stellar nuisances, external environment	galaxy-level likelihood with selection and covariance

5 Diagnostic Results

The reported CMB overlays, compressed background comparisons, and galaxy scaling checks are diagnostic. Their formulas or inputs are selected with knowledge of the comparison data, omit required covariance or nuisance structure, or lack a physical source map. They test numerical paths and conditional formulas and do not establish a cosmological prediction.

6 Retrospective Comparison Ledger

A deterministic ledger released with these papers records every comparison of a conditional cosmological value of the construction with public data. Its producer reads the corpus values and the parent receipts, types each public measurement once with its citation, and writes one row per comparison with a class label; an independent verifier recomputes every row. Every row is on seen data. No row is a frozen prediction or a score, and the rule, theory minus measurement over the quoted standard deviation, is fixed before any row is read. Table 1 summarizes the rows.

The rows that agree are, with three exceptions, shared with the standard baseline and support

Quantity	Value	Public measurement and distance	Class
n_s	$1 - P_\star/48 = 0.96602$	Planck 2018 0.9649 ± 0.0042 ($+0.27\sigma$); SPT-3G with Planck 0.9636 ± 0.0035 ($+0.69\sigma$); SPT, Planck and ACT 0.9684 ± 0.0030 (-0.79σ); ACT DR6 with Planck 0.9709 ± 0.0038 (-1.28σ); with lensing and BAO 0.9743 ± 0.0034 (-2.43σ)	conditional theorem; discriminates
$\mathrm{d}n_s/\mathrm{d}\ln k$	0	Planck 2018 -0.0045 ± 0.0067 ($+0.67\sigma$); ACT DR6 with Planck, lensing and BAO 0.0062 ± 0.0052 (-1.19σ)	conditional theorem; discriminates only against running of order 10^{-2}
r	0	BICEP/Keck 2018 $r_{0.05} < 0.036$; the fourth Planck release with BK18 and BAO $r < 0.032$; consistent	conditional theorem; discriminates against generic slow roll
β_{iso}	0	Planck 2018 uncorrelated CDI $100\beta_{\mathrm{iso}} < 3.5,\ 3.8,\ 3.9$ at the three reference scales; consistent	shared baseline
f_{NL}	0 at the source	Planck 2018 local -0.9 ± 5.1 ($+0.18\sigma$), equilateral -26 ± 47, orthogonal -38 ± 24	shared baseline
Ω_K	0	Planck 2018 with lensing -0.0106 ± 0.0065 ($+1.63\sigma$); with BAO 0.0007 ± 0.0019 (-0.37σ); DESI DR2 with CMB 0.0023 ± 0.0011 (-2.09σ)	shared baseline
(w_0, w_a)	$(-1, 0)$	DESI DR2 BAO with CMB 2.61σ; with Pantheon+ 2.60σ; with Union3 3.38σ; with DESY5 3.93σ (Gaussian moment diagnostic of the official chains)	conditional theorem; tension
$\Lambda\ell_P^2$	$3\pi/N$, $N = 3.29$ and 3.30×10^{122}	Planck-central Λ: $+0.63$ and $+0.38$ percent; DESI DR2 with CMB display $(2.968 \pm 0.040) \times 10^{-122}$: -3.5 and -3.8 percent (-2.6 and -2.8σ)	closure-candidate display
a_0	no source value	SPARC calibration $1.161 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at mass-to-light 0.5; ratio to $c^2\sqrt{\Lambda/3}$ equal to 0.214	fitted comparison value

Table 1: Rows of the cosmology postdiction ledger. Distances are theory minus measurement in units of the quoted standard deviation; bounds are consistent when the value lies below the bound. All data are seen.

nothing: zero isocurvature, zero non-Gaussianity at the source and zero curvature are the single-field adiabatic expectation, and the fixed-capacity point is the ΛCDM null. The exceptions are the exact tensor zero, which generic slow-roll inflation does not share and which a null at the LiteBIRD design sensitivity would discriminate, the exact zero running, which discriminates only against running of order 10^{-2}, and the specific tilt value, which the seen combinations place between $+0.7\sigma$ and -2.4σ. The tensions are the fixed-capacity equation of state, at 2.6 to 3.9σ across the DESI DR2 combinations under the moment diagnostic and with monotone-branch posterior fractions between 0.007 and 0.11 percent, the curvature value from DESI DR2 with CMB at 2.1σ, and the capacity display against the DESI DR2 Λ coordinate at 2.7σ, while the same candidates sit within half a percent of the Planck-central value. These are retrospective diagnostics of receipt-conditional values. The freeze proposals for the tilt, the tensor zero, and the running and isocurvature zeros, and the pending fixed-capacity row, define the only route from this ledger to a score: a frozen combination, statistic and thresholds fixed before the eligible release.

7 No-Data-Use Boundary

A source-derived calculation must expose its dependency graph. The source side may use mathematical constants, declared OPH finite artifacts, and explicitly imported physical inputs. It may not use the observed quantity being claimed as an output, a fit to that quantity, or a proxy calibrated from the same dataset.

This provenance rule is enforced through the declared dependency graph and data-use classification.

8 Theorem scope

The screen-spectrum and radial-lift mathematics are conditional theorems. One-shell inversion is nonunique; physical source dilation and radial cross-covariance tomography are two established sufficient uniqueness routes. Finite source instantiation, repair-charge cosmological source, Boltzmann bridge, and joint likelihood are not supplied. All cosmological data products are diagnostic. No cosmological result carries an OPH falsification verdict.

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